

Functional Localization and Cortical Architecture in the Nine-banded Armadillo (*Dasypus novemcinctus mexicanus*)

G. JAMES ROYCE,^{1,2} GEORGE F. MARTIN,³ AND RICHARD M. DOM⁴

¹ Department of Anatomy, Albany Medical College, Albany, New York;

² Department of Anatomy, Bardeen Medical Laboratories, University of Wisconsin, Madison, Wisconsin; ³ Department of Anatomy, The Ohio State University, Columbus, Ohio; ⁴ Department of Anatomy, Medical University of South Carolina, Charleston, South Carolina

ABSTRACT A functional map of the armadillo neocortex was produced by cortical stimulation and recording evoked potentials following somatic, auditory and visual stimuli. The results obtained were then correlated with the cortical architecture as revealed by Nissl, Golgi and myelin-stained sections. Cortex rostral to the supraorbital sulcus has a wide layer IV and is mostly silent, except for a motor eye field and a part of the tongue sensory region in its caudal part. Two types of motor-sensory cortex are present caudal to the supraorbital sulcus. Postsupraorbital I is mostly motor and has prominent pyramidal layers. Layer V is particularly well developed and in rostral sections its superficial zone is broken up into clusters similar to the solid "barrels" seen in layer IV of other species. Postsupraorbital II has less prominent pyramidal layers and layers II and III are organized into clusters. This region corresponds to the sensory area for the limbs and trunk and the partially overlapping (surface recordings) sensory and motor areas for head, snout and tongue. Digits and limbs are rostral to the trunk representation in both the sensory and motor "homunculi." Even though surface recording was employed, potentials evoked by visual stimuli could only be recorded from a small caudal area with a very thin layer IV. Although striate and peristriate areas appear similar in Nissl stained preparations, they can be readily differentiated in Weil stained sections. The stellate character of neurons in layer IV of the visual cortex is particularly apparent in Golgi material. Auditory evoked surface potentials were recorded from a broad oval region in the caudal lateral cortex which has a wide layer IV and aggregates of neurons in layers II and III. A Weil stain demonstrates inner and outer bands of Baillarger in this same region. The presumptive insular cortex is electrically silent to sensory stimulation and presents as a narrow band just dorsal to the rhinal fissure with indefinite cell lamination and little myelin.

The brain of *Edentata* has received little attention since the appearance of the classical monograph on this subject by G. E. Smith in 1899. Most of the studies which have appeared since then on the armadillo neocortex have dealt with corticofugal fiber pathways (Fisher et al., '69; Strominger, '69; Harting and Martin, '70 a,b; Dom et al., '71), although the latter study included a brief description of the histology of the cortex adjacent to the supraorbital sulcus and neurophysiological data on the motor area as defined by electrical stimulation.

Although many external features of the nine-banded armadillo (*Dasypus novem-*

cinctus mexicanus) (fig. 1) are very specialized (the elongated snout and tongue, and the unique armor or "carapace"), most of its central nervous system, particularly its relatively small neocortex (figs. 2, 3), appears to be of a generalized nature. We have attempted to map the cortical sensory representation in this species by evoked potential techniques (and hence to compare these to the motor regions as defined by Dom et al., '71) and to correlate such maps

² Reprint requests should be sent to the present address of the first author which is: G. James Royce, Department of Anatomy, Bardeen Medical Laboratories, The University of Wisconsin, Madison, Wisconsin 53706.

with the cortical architecture revealed by the Nissl, Weil and Golgi techniques. It is our intent that the information reported herein will provide a comparative reference for cortical organization in the order *Edentata*.

MATERIALS AND METHODS

Twelve adult armadillos (fig. 1), each weighing between three and five kilograms, were utilized for the evoked potential portion of this study. The first two animals were anesthetized initially with chloralose (60 mg/kg I.P.) followed by sodium thiopental (15 mg/kg I.V.) which was administered just prior to the surgical procedure to suppress reflex contractions. The sodium thiopental was discontinued after surgical manipulation and the animal was allowed to stabilize under the influence of chloralose for 60 minutes. Although positive results were obtained with this method, the occurrence of spontaneous or reflexly induced movements dictated the use of either a blocking agent or another anesthetic for maximal utilization of the acute preparation. Therefore, sodium pentobarbital (30 mg/kg I.V.) was the anesthetic agent employed in the remaining ten experiments.

In order to have fixed reference points for the somatosensory recordings the anesthetized animal was positioned in a Stoelting stereotactic device for stabilization of the head. For the visual and auditory stimulation, the head was immobilized but not positioned in the stereotactic apparatus. After positioning the head the skin was incised, the epidermal plate severed, and the calvarium removed over one cerebral hemisphere. Reduction of pulsations was effected by draining the cisterns ambiens and magna of cerebrospinal fluid, and subsequently by the application to the exposed cortex of a preparation of 3% agar in an 8.5% sucrose solution. Also, Decadron phosphate (Merck & Co., 2 mg/kg I.V.) was utilized to help prevent edema of the exposed brain.

To establish the somatic representation on the surface of the cerebral cortex, a variety of stimulation techniques were utilized. For example, the hairs were mechanically displaced, the joints rotated or an electrical stimulus having a duration of 1 ms or less was applied to the skin or an exposed cutaneous nerve. The armadillos that

were used to record the special senses were doubly utilized, i.e., the same animal was used for the recording of potentials evoked by both auditory and visual stimuli. In all cases the visual area was plotted initially and after this was completed, the auditory cortex mapped. In mapping the visual cortex, non-discrete (whole field) stimulation of the retina was obtained with a photoflash and photometer screen in conjunction with pupillary dilation (one or two drops of a commercial solution of cyclopentolate hydrochloride). The auditory area was mapped using the clicking noise of a camera shutter as the stimulus.

Two types of electrodes were used for recording the evoked surface potentials. Monopolar silver-silver chloride and stainless steel electrodes were used to obtain a rough pattern of sensory localization. Subsequent explorations were carried out by using concentric bipolar electrodes for a more precise mapping of evoked potentials. The signal detected by the cortical electrode was preamplified before being displayed on the screen of a cathode ray oscilloscope (Tektronix 565 or 502A) and subsequently photographed. In some cases, a storage type

Abbreviations

A.	Abdomen
B.	Arm (brachium)
C.	Leg (crus)
D.	Back (dorsum)
D ₁ .	Digits of the hand (manus)
D ₂ .	Digits of the foot (pes)
F.	Forearm (antebrachium)
H.	Buttock (nates)
IN.	Insular cortex
L.	Lower lip (labium mandibulare and suprahyoid portion of neck)
N.	Neck (ventral portion)
OB.	Olfactory bulb
P.	Pectoral region (pectus)
POSTSPR I.	Postsupraorbital I
POSTSPR II.	Postsupraorbital II
PRS.	Presupraorbital cortex
PS.	Peristriate cortex
PST I.	Postsupraorbital I
PYR.	Pyriiform cortex
RF.	Rhinal fissure
RS.	Retrosplenial area
S.	Snout (includes labium maxillare)
SG.	Sagittal sulcus
SO.	Supraorbital sulcus
Sp.	Scalp
SS.	Suprasylvian sulcus
Sv.	Striate or visual cortex
T.	Tongue
Ta.	Temporal or auditory cortex
Th.	Thigh (femur and genu)
Tl.	Tail

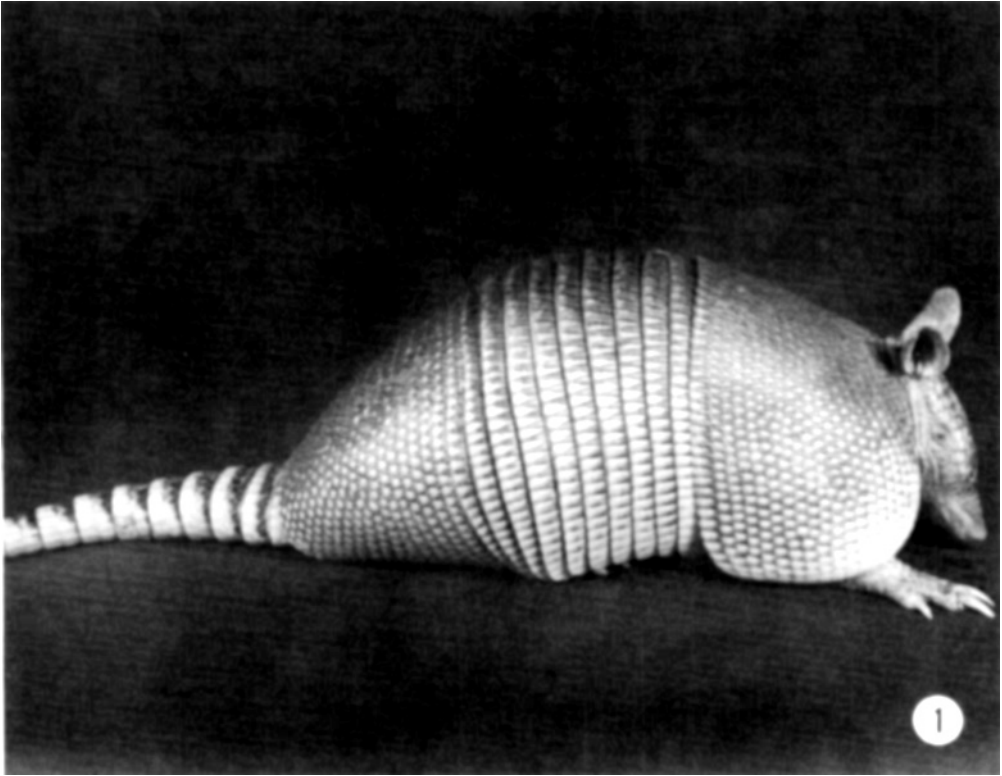


Fig. 1 Photograph of the nine-banded armadillo, *Dasypus novemcinctus mexicanus*. Note the small eyes and large ears, as well as the bony carapace with nine bands which covers the body. There is very little hair present on the exposed surfaces of this species.

oscilloscope was used for editing purposes.

Fourteen specimens were used for the anatomical portion of this study. The nine animals used for cytoarchitecture and myeloarchitecture were anesthetized with sodium pentobarbital and perfused intracardially with saline followed by 10% formalin. The brains were removed and allowed to fix for periods of time varying from one week to over a year. Following fixation the brains were cut on a freezing microtome at thicknesses between 20–50 micra in either the frontal or parasagittal plane, or tangential to the neocortical surface, and stained by either the thionin, cresyl violet, or Weil methods. The remaining five brains were processed by variations of the Golgi method, which included the rapid Golgi technique (Morest and Morest, '66), the procedure of Fox et al. ('51) and the Golgi-Cox methods (Sholl, '53; Ramon-Moliner, '70). Cortical thicknesses were measured directly by the

use of an American Optical Company Micrometer slide.

RESULTS

Data obtained as a consequence of this study fall into two categories: evoked potential responses recorded from the cortex, and the histological characterization of cortical regions. The reader is referred to figures 2 and 3 for lateral and dorsal views of the armadillo cerebral hemispheres. Following the terminology of G. Eliot Smith (1899) the supraorbital (SO), sagittal (SG) and suprasylvian (SS) sulci are labelled as well as the rhinal fissure (RF) and will be referred to in the following account.

Results of evoked potential studies

Figure 4 is a composite of the observations obtained from 12 animals. Stimulation of the tongue evoked potentials from a region slightly rostral to the supraorbital

sulcus, although most of the tongue responses were obtained from the cortex immediately caudal to the supraorbital sulcus, but ventral to Smith's sagittal sulcus. Evoked potentials from the snout (S) were recorded from a region immediately dorsal to the rhinal fissure, but caudal to the tongue area. Responses to stimulation of the

lower lip (L), neck (N), and scalp (Sp) were elicited dorsal to the snout area, but were confined to the region between the sagittal and suprasylvian depressions.

Responses from stimulation of the rostral limb were obtained from a region dorsal to the somatic sensory areas just described. In a rostral to caudal sequence evoked po-

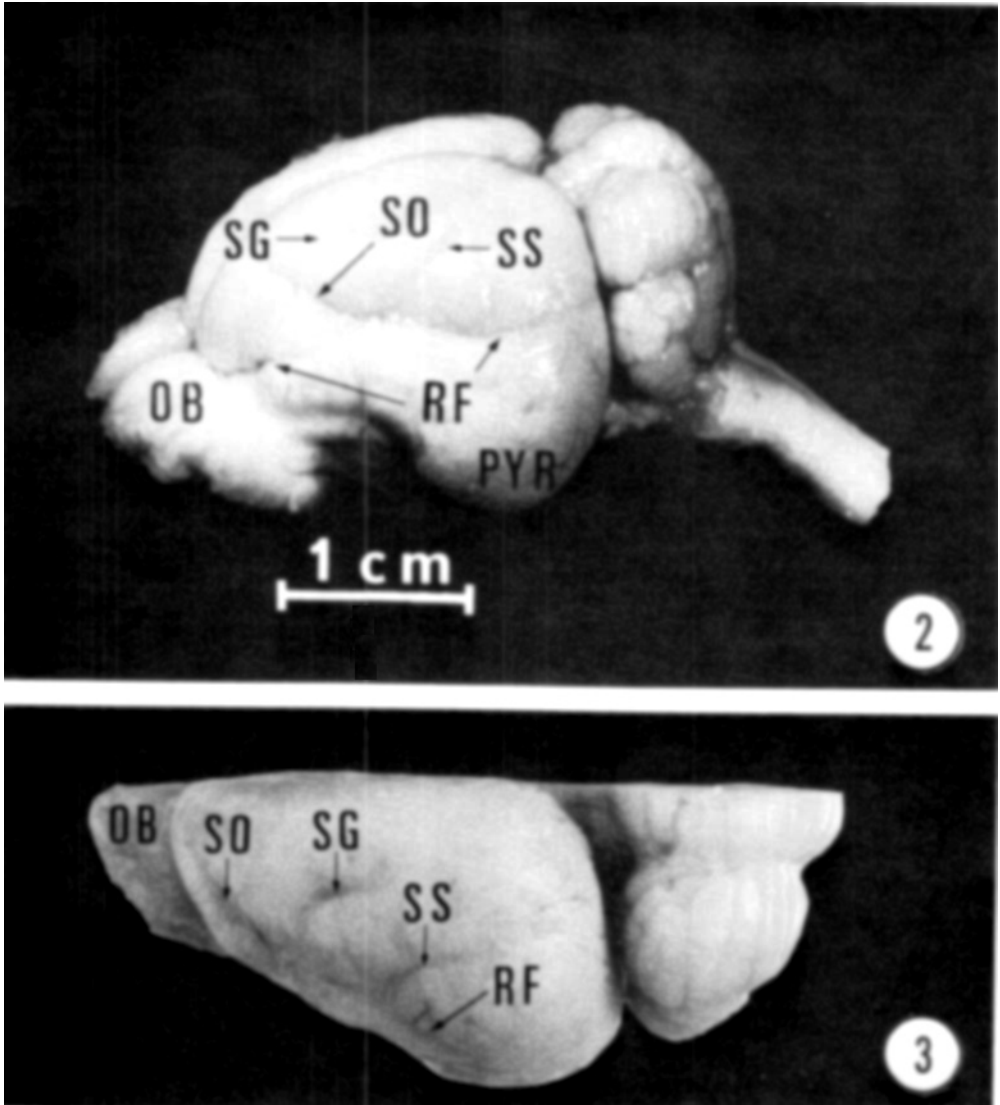
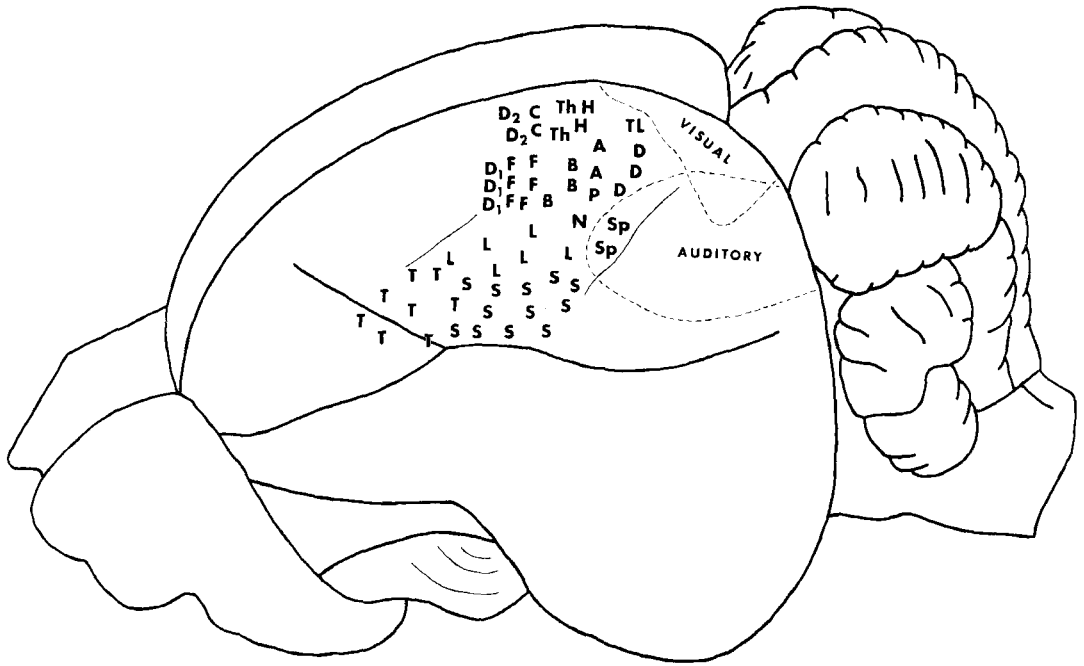


Fig. 2 Photograph of the lateral aspect of the armadillo brain. The sagittal sulcus (SG), the supraorbital sulcus (SO), the suprasylvian sulcus (SS), the olfactory bulb (OB), the rhinal fissure (RF) and the pyriform cortex (PYR) are labeled for reference.

Fig. 3 Photograph of the dorsal aspect of one hemisphere of the armadillo brain. The olfactory bulb (OB), the supraorbital sulcus (SO), the sagittal sulcus (SG), the suprasylvian sulcus (SS) and the rhinal fissure (RF) are labeled for reference.



4

Fig. 4 Drawing of a lateral view of the armadillo brain. The evoked potential results of somatic sensory stimulation are indicated as follows: A, abdomen; B, arm (brachium); C, leg (crus); D, back (dorsum); D₁, digits of the hand (manus); D₂, digits of the foot (pes); F, forearm (antebrachium); H, buttock (nates); L, lower lip (labium mandibulare and suprahyoid portion of neck); N, neck (ventral portion); P, pectoral region (pectus); S, snout (includes labium maxillare); Sp, scalp; T, tongue; Th, thigh (femur and genu); and TL, tail. The visual and auditory areas are indicated by dashed lines.

tentials were elicited from the digits of the "hand" (D₁), forearm (F), and arm (B), respectively. Responses were recorded from the abdomen (A), pectoral region (P) and back (D) in the area caudal to sensory region of the forelimb.

The somatic sensory area of the hindlimb was represented most dorsally on the dorso-lateral surface of the hemisphere. In a pattern similar to that of the forelimb, the hindlimb responses were elicited in rostral to caudal sequence from the foot (D₂), leg (C), thigh (Th) and buttock (H). Responses from the tail (TL) were from an area caudal to the buttock region and dorsal to the back representation.

The auditory area occupied a broad oval region in the caudal and lateral portions of the neocortex. In its more rostral portion, the auditory region was considerably dorsal to the rhinal fissure, but gradually ap-

proached this fissure toward the caudal pole of the hemisphere. Most auditory responses were elicited caudal to the suprasylvian sulcus, although some could be obtained more rostrally in a region which overlapped the somatic sensory areas of the scalp and back.

The relatively small region from which visual evoked potentials were obtained had an irregular configuration and was located on the most dorsal and caudal portion of the hemisphere. Rostrally, the visual region abutted upon, but did not overlap the sensory areas of the buttocks and tail. Ventrally, there was a triangularly-shaped extension of the visual area which overlapped the dorsal part of the auditory representation.

There were neocortical regions which yielded no evoked potentials from any form of stimulation. Evoked potentials could not be obtained from most of the presupraorbi-

tal cortex — except for a small area close to the ventrolateral part of the supraorbital sulcus where tongue responses were elicited. All of this region is not "silent" however, because eye movements were obtained from a large region dorsal to the tongue sensory area (Dom et al., '71). Although most of the region dorsal to the supraorbital sulcus and rostral dorsal to the sagittal sulcus yielded no evoked potentials, this area corresponded to that previously described where forelimb and trunk movements were elicited upon electrical stimulation (Dom et al., '71). However, "silent" cortex does appear to be present between the ventral border of the auditory area and the rhinal fissure where neither motor responses (Dom et al., '71), nor evoked potentials could be recorded. This silent area included the insular cortex, as defined histologically (RESULTS *Cortical architecture*), but was much broader than that, particularly in its dorsal region, where it abutted upon the sensory scalp area. Although the entire lateral neocortex was explored with electrodes, no evidence was found for secondary sensory areas.

The table below presents the slow wave evoked potential results in terms of amplitude and latency. The surface response was always initially positive and the amplitude of the positive response is shown in the table. It is evident that the auditory (as high as 1.2 mv) and somatic responses (as high as 1.0 mv) were roughly similar in amplitude, but at least twice as great as those evoked by visual stimulation (as high as 0.5 mv). Also the mean latency of the visual response (30 ms) was at least three times that of either the mean auditory (10 ms) or the mean somatic response (7.5 ms).

Table of slow wave evoked potentials

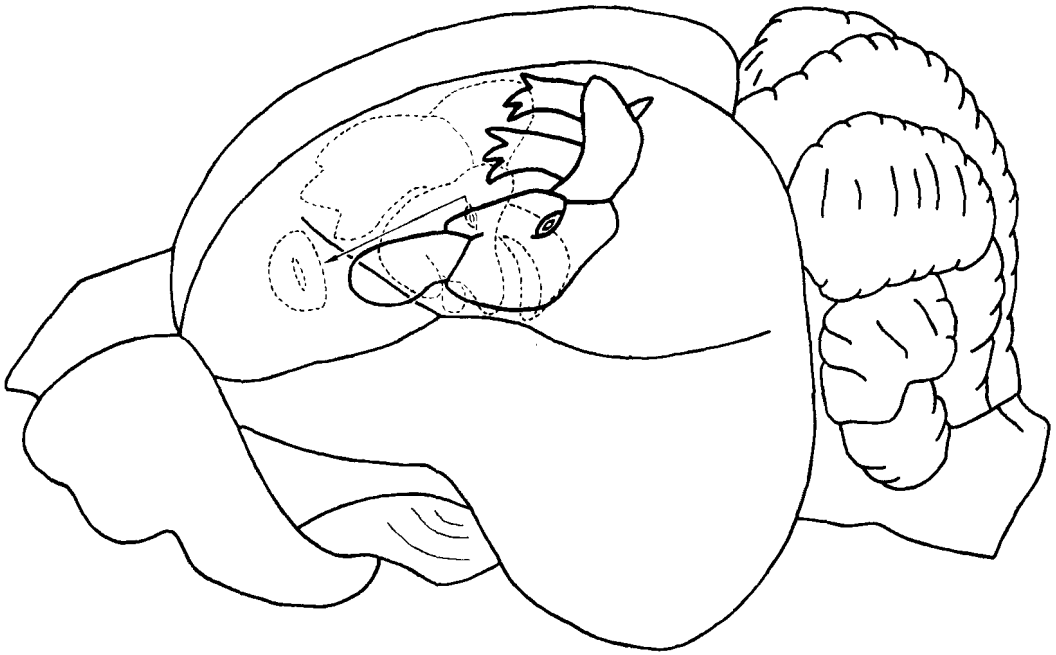
	Amplitude	Mean latency
<i>Response</i>		
Auditory	as high as 1.2 mv	10 ms
Visual	as high as 0.5 mv	30 ms
Somatic	as high as 1.0 mv	7.5 ms

The evoked potential results (fig. 4) are presented in a homunculus type drawing (fig. 5) in which the sensory representation is shown in solid heavy lines. It is obvious that the head and tongue representations

are disproportionately large and are directed rostrally, as are those of the limbs. The results of a previous study (Dom et al., '71) formed the basis for drawing the motor homunculus shown in figure 5 (thin, broken lines). The appendage emanating from the right side of the motor head region outlines the large area from which movements of the pinna of the ear were obtained. Rostral to the motor nose region is the area for the upper and lower lips, between which tongue movements could be induced. It may be noted that the sensory tongue representation is much larger than the motor area for the same region. Artistic license was taken in placing the motor eye area in the head region and the fact that it is not located there is shown by the arrow which points to the motor eye field found rostral to the supraorbital sulcus. Dorsal to the motor head region is the rather large area from which movements of the forelimb (rostrally) and trunk (caudally) were elicited. Movements of the hind limb were obtained on the medial surface of the hemisphere and are not shown in the drawing, i.e., although their orientation was similar to that of the forelimb (i.e., digits are represented rostral to the hip). Thus the digits are facing rostrally in both the sensory and motor homunculi, and the motor and sensory representations appear to be rotated with respect to one another at an angle of 25 degrees or more, with the axis of rotation roughly at the base of the tongue.

Cortical architecture

For purposes of reconstruction, the neocortex was examined microscopically and compared to gross landmarks. Useful landmarks included paleocortex, archicortex and olfactory related structures, and various portions of the diencephalon. Two structures of the pallium which were consistent and particularly useful for reconstruction were the rhinal fissure, (which frequently appeared to be discontinuous grossly, forming the anterior and posterior rhinal fissures as described by Smith, (1899) and the supraorbital sulcus (β of Smith, 1899) (fig. 2). Also present, but variable in location was the suprasylvian sulcus (δ) of Smith and more variable yet, a slight depression dorsal to the supraorbital sulcus which appears comparable to Smith's sagittal (γ) sulcus (fig. 3). Although frequently



5

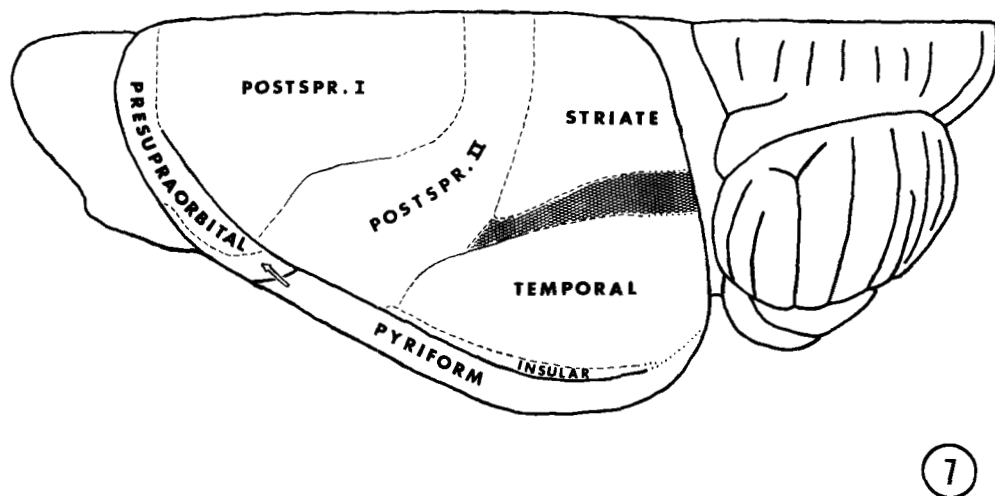
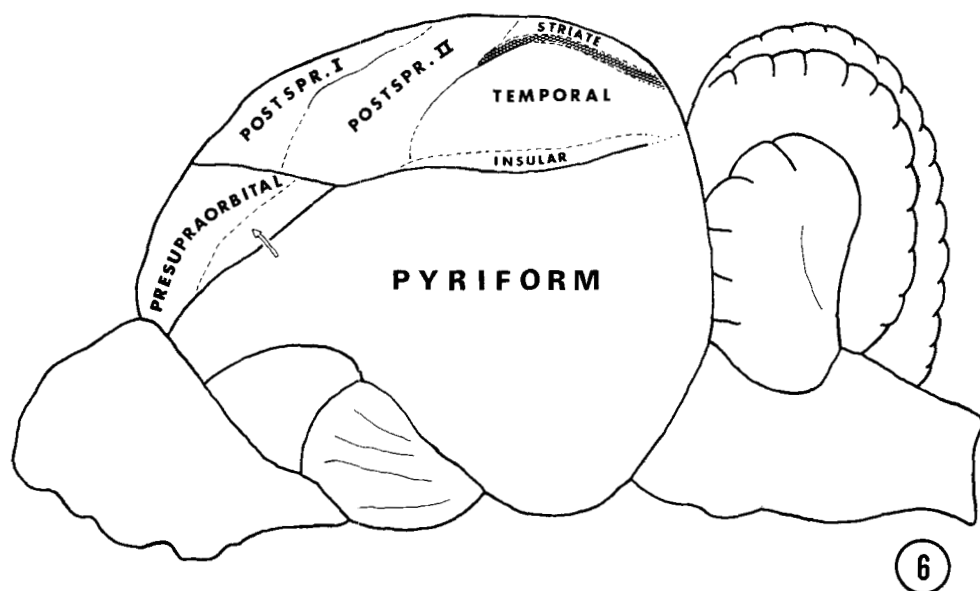
Fig. 5 Drawing of the lateral aspect of the armadillo brain. The sensory homunculus is shown in solid heavy lines. The motor homunculus (from Dom et al., '71) is represented by dashed lines.

indistinct grossly, these latter two sulci could usually be seen microscopically and proved to separate histologically as well as functionally distinct areas. For this reason the cytoarchitectural areas described in this account have been named in reference to the above fissures and sulci. Similar nomenclature has been used previously for both the opossum (Gray, '24; Walsh and Ebner, '70), and armadillo (Dom et al., '71), and was selected in the present study in preference to ascribing numbers corresponding to areas of Broadmann ('09). For example, the neocortex rostral to the supra-orbital sulcus is referred to as presupra-orbital and areas caudal to but near the same sulcus have been designated as post-supra-orbital.

In the most rostral regions of the pallium the rhinal fissure serves as a clear demarcation between neocortex and pyriform cortex. The most obvious difference between these two regions is the larger number of neurons in the external granular layer of

the pyriform cortex. However, in more caudal regions this densely packed external granular layer sweeps above the rhinal fissure (fig. 8). Thus, in reconstructing this region a small portion of the cortex dorsal to a depression that is continuous with the rhinal fissure has been designated as pyriform cortex (figs. 6, 7).

Presupraorbital pallium which is clearly neocortical is relatively thick varying from 4.1 mm rostrally to 3.4 mm caudally. In the presupraorbital area (figs. 8, 9) layer I is fairly wide (0.3 mm to 0.4 mm) and relatively cell-free. Layer II is a wide band of small granule cells whereas layer III consists of many small pyramidal cells. Layer IV is quite wide (1.1 mm) and consists of many deeply staining round or oblong neurons intermingled with a lesser number of small pyramidal and multiform cells. Layer V is a discontinuous band of small pyramidal cells and layer VI consists mostly of very small lightly-staining round or fusiform cells.



Figs. 6, 7 Drawings of the lateral (top) and dorsal (bottom) aspects of the armadillo brain with cytoarchitectural areas as determined from reconstructions. Postspr. I and II are abbreviations for postsupraorbital I and II. The stippled area demarcates the limits of the peristriate cortex. The sulci and fissures are in solid lines and are the same as shown in figures 2, 3. The cytoarchitectural areas are shown in dashed lines with uncertain boundaries indicated by dotted lines.

As mentioned previously, there is a slight oblique depression in the postsupraorbital region (fig. 3) which is variable in location and prominence, and is probably the sagittal (γ) sulcus of Smith (1899). This sulcus

is usually visible in histological sections and when present appears to separate two different cortical regions: postsupraorbital I (dorsally), and postsupraorbital II (ventrally).

The histology of postsupraorbital I changes from rostral to caudal, but not so drastically as to warrant dividing it into two portions (figs. 6, 7, 8, 10, 11, 12, 13). Postsupraorbital I cortex averages 2.7 mm in thickness, but is somewhat thicker in its dorsal and rostral portions. Layer I is fairly wide (0.35 mm to 0.40 mm), and contains a few scattered cells; layer II is relatively thin consisting of aggregates of small granule cells; layer III is a distinct band of small pyramidal cells and is much thicker than layer II. Layer IV of postsupraorbital I is a continuous band of small and medium-sized granule cells. Layer V is particularly well developed in this region and caudally, it is divisible into two sublayers (fig. 13): a superficial zone of small pyramidal cells (Va), and a deeper region (Vb) of larger more intensely stained pyramidal cells possessing apical dendrites which can be followed only for a short distance in Nissl stained sections. These dendrites can be better visualized in Golgi preparations (fig. 14). In a low-power view of a Golgi-Cox impregnation of postsupraorbital I (fig. 16) it can be seen that there are two distinct layers of pyramidal cells with long apical dendrites and extensive basal dendritic arborizations. These layers correspond to layers III and V as seen in the thionin sections.

In the rostral regions of postsupraorbital I the two divisions of layer V are still present, but the pyramidal neurons in the deeper zone (Vb) are smaller and stain less intensely (figs. 8, 10). The neurons in the more superficial zone (Va) are organized into clusters (figs. 8, 10, 11) which are roughly ovoid and are surrounded by regions of diminished cellularity. The clusters extend onto the medial neocortex in the region immediately adjacent to the supraorbital sulcus, but in slightly more caudal regions they are present only in the dorsal cortex. In sections cut tangentially to the cortical surface the cell clusters of layer V could be faintly visualized at a very low power, but did not lend themselves well to photography. They appeared as solid, round cell clumps, surrounded by less cellular septa, and thus must be considered to be cylindrically shaped aggregations of neurons.

The cortex of postsupraorbital II is striking at low power due to the scalloped appearance of its lateral portion (fig. 12), pro-

duced by the regularly-spaced cell clusters of layers II and III which are indented by layer I. Thus, layer I varies in thickness from 0.1 mm to 0.4 mm. Layer IV (fig. 15) is present, but thin. Although layer V is obvious, the pyramidal neurons are smaller and reduced in number when compared to cells of this layer in postsupraorbital I and layer V is not divisible into two zones. In postsupraorbital II layer VI consists of many round and fusiform cells intermingled with the smaller pyramidal cells of the deeper portions of layer V. At its maximum dimension, layer VI is relatively, very wide (up to 1.7 mm). The total thickness of the cortex of this region is 2.7 mm. In a low-power view of a Golgi preparation, the pyramidal neurons are smaller, have less extensive basal dendritic trees and shorter apical dendrites than those in postsupraorbital I (fig. 16).

Cortex in the visual area (fig. 17, 19) designated as striate, is thin (2.3 mm), and has a wide layer I. Layers II and III appear to be fused, consisting of an admixture of small granular and pyramidal neurons which are frequently organized into small clusters. Layer IV is not well developed and its depth is difficult to determine in thionin stained sections. The portion which consists strictly of small granule cells measures only 0.4 mm, but granule cells are intermingled with pyramidal neurons in the more superficial zone of layer V and stellate cells appear in Golgi preparations to a depth of 0.6 mm (fig. 21). Layer V is a distinct layer of pyramidal cells, the more superficial of which blend with layer IV. Layer VI is relatively wide and consists mostly of fusiform cells (fig. 19).

Weil stained sections (fig. 22), show myelinated fibers in the more superficial cortical layers of the visual area. The region of these fibers ends abruptly laterally. The number of fibers in the superficial layers increases sharply in the auditory region. Upon examination of serial sections this region of diminished myelinization between the visual and auditory regions proved to be a continuation of the cortex dorsal to the suprasylvian sulcus (figs. 6, 7). In Nissl stained material this small region differs little from the striate area except for a more uniform layer I (fig. 20), and a lack of cell clusters in layers II and III. Because of its position it has been designated as lateral peristriate cortex. It is not clear whether

there is any medial peristriate cortex. Near the dorsal midline the retrosplenial cortex appears abruptly having a markedly increased number of small granule cells in layer II (fig. 17). A slight sulcus is variably present between the striate cortex and the retrosplenial area which may be sulcus (α) as described by Smith (1899). When this sulcus can be visualized there is a very small transitional region which may be a type of medial peristriate cortex, but the identification of this is uncertain.

The auditory region, designated temporal cortex, is caudal to the suprasylvian sulcus (fig. 6, 7, 17) and is thin (2.5 mm). It has a relatively thin layer I (0.25 mm) although "extensions of layer I" penetrate as far as the upper part of layer IV and separate clusters of cells in layers II and III at fairly regular intervals (fig. 18). In the temporal area, there are more granule cells in layer II, and the number of pyramidal cells in layer III is greatly reduced when compared to the striate area. Layer IV is relatively thick (0.8 mm) and consists mostly of granule cells larger than those in layer IV of the striate area. Layer V is an indistinct band of pyramidal cells and layer VI consists of oblong or flattened cells intermixed with small triangular neurons. The temporal region, as seen in Weil stained material (fig. 22), contains an inner and outer band of Baillarger and, in some sections a peculiar band of myelinated fibers in layer II (fig. 23).

There is a small, myelin poor area lateral to the temporal region and adjacent to the rhinal fissure (fig. 22). Thionin stained sections of this region show that it becomes thin as it approaches the rhinal fissure (2.0 mm to 1.5 mm at the rhinal fissure) and that it is characterized by a diminution in size of all the cortical layers, but particularly layer IV which is absent at the rhinal fissure (fig. 24). Because of its position, this area has been designated as insular cortex.

DISCUSSION

In this study somatic sensory, visual, and auditory evoked potentials were obtained in specific regions of the armadillo neocortex. Cytoarchitectural and myeloarchitectural analyses, supplemented to a limited extent by Golgi material afforded the basis for producing maps of histologically different cortical regions. Since surface record-

ings were used in order to obtain an early overview of cortical localization the areas defined may be considerably smaller than reported herein and perhaps conform even more closely to the anatomical boundaries. The question of correspondence between cytoarchitecturally defined borders and functionally distinct areas in the armadillo cortex is not directly addressed by our results and deserves further study by intracortical recording techniques. It is the purpose of this section to discuss these regions in rostral to caudal sequence, to correlate them with the sensory and motor regions as delineated physiologically, and to compare these results with the pertinent literature.

Much of the cortex rostral to the supra-orbital sulcus did not respond to either the stimuli used in this study or to electrical stimulation (Dom et al., '71) and, therefore, may be considered "silent." The exceptions to this are located in proximity to the supra-orbital sulcus, where the motor eye fields and part of the sensory tongue representation were found. There is a small wedge-shaped area of ventral, caudal presupra-orbital cortex which is histologically similar to pyriform cortex and would thus be expected to be "silent." The shallow depression in this region thus may not be the rhinal fissure. In fact, Smith (1899) did not recognize a continuous rhinal fissure in this species, but rather designated anterior and posterior rhinal fissures between which was a gap in his drawing; the gap corresponds to the region containing pyriform cortex. There is, however, a relatively large area of "silent" presupraorbital cortex which is clearly neocortical. This is in contrast to some reports of the relative lack of such areas in the opossum and the hedgehog. In the opossum eye movements were elicited in most, if not all, of the frontal cortex rostral to the sensorimotor field (Lende, '63b), and in the hedgehog the back, tongue and eye motor fields occupied the entire frontal region (Lende and Sadler, '67). However, cortex of the "prefrontal" or "orbitofrontal" type is not necessarily absent in the opossum if Rose and Woolsey's ('48) definition of such cortex is accepted. They defined orbitofrontal cortex as that area which receives essential projections from the nucleus medialis dorsalis. This nucleus has been shown to have a dis-

tinct system of frontal projections in the opossum (Bodian, '42) and lesions in the frontal and preorbital areas in this species result in both orthograde (Martin and Fisher, '68; Martin, '68) and retrograde (Pulbols, '68) degeneration in the dorsomedial nucleus. The question of whether or not the presence of connections with the dorsomedial nucleus of the thalamus determine that such cortex is similar to prefrontal neocortex in primates is discussed at length in another paper (Voneida and Royce, '74).

In the armadillo it has been demonstrated that terminal degeneration in the dorsomedial nucleus results from lesions in rostral as well as caudal portions of presupraorbital cortex (Harting and Martin, '70a). This would suggest that at least some portion of this region in the armadillo is similar to the prefrontal cortex of other species, although such a conclusion would not be drawn from its cytoarchitecture. In the monkey, for example, the frontal granular cortex "is characterized by shallow depth, lack of special differentiation, and the presence of a clearcut, but narrow, layer IV with granular cells," and in Carnivora it is said that in this cortex "the fourth layer is distinguishable as a lightly stained band, indicating that it is sparsely populated with cells," (Akert, '64). In the armadillo, however, we have confirmed what has been shown previously (Dom et al., '71), that layer IV of the presupraorbital area is composed of a broad band of granular neurons. Also, this cortex is relatively thick (3.4–4.1 mm) when compared to other cortical regions (striate, for example is 2.3 mm). Such results make it apparent that cross species homologies can not be arrived at purely on the basis of cytoarchitecture, but must be approached by a multiplicity of techniques. There is the possibility that in the armadillo the development of the presupraorbital granular layer reflects some type of olfactory specialization (an hypothesis suggested by the enormous development of its olfactory structures) and that such cortex has some function in olfactory association.

As mentioned previously, a motor eye field has been demonstrated in the caudal presupraorbital region of the armadillo cortex (Dom et al., '71), a finding which may correlate with the existence of projections from this region to the ventral nucleus of the periaqueductal gray which lies in prox-

imity to the oculomotor and trochlear nuclei (Harting and Martin, '70). The small portion of caudal presupraorbital cortex in which evoked potentials from the tongue were elicited does not differ either cytoarchitecturally or by demonstrated connections from other neocortex in this region.

A large portion of the region caudal to the presupraorbital sulcus contains two different kinds of cortex which have been designated postsupraorbital I and II. These two cortical regions are partially and inconsistently separated by a shallow sulcus, the sagittal (γ) sulcus of Smith (1899) and differ in two main respects: (1) postsupraorbital I has a thick layer V divisible into two zones, whereas this layer is smaller and consists of only one zone in postsupraorbital II; (2) and the external configuration of layers II and III is smooth in postsupraorbital I, whereas these layers are broken up into clusters in postsupraorbital II. This latter feature imparts a "scalloped" appearance to this cortex at low power.

Postsupraorbital I cortex of the armadillo may be compared to similar regions in other species. In the postorbital cortex of the opossum pyramidal neurons predominate and layer IV is either very thin or absent (Gray, '24; Walsh and Ebner, '70). This region in the opossum has been shown to give origin to some fibers of the corticospinal tract (Martin and Fisher, '68) and to produce motor responses upon electrical stimulation (Lende, '63b). Likewise, postsupraorbital I cortex of the armadillo has been shown to give origin to fibers of the corticospinal tract (Dom et al., '71) and motor responses are elicited from electrical stimulation of this area (Dom et al., '71). However, this region differs from the postorbital cortex of the opossum in that it has a distinct layer IV and stellate cells are obvious in Golgi preparations. It should be noted, however, that the postorbital area is small in the opossum and that the adjacent parietal cortex (which contains a well developed internal granular layer) is also included in the motor representation. The armadillo postsupraorbital I cortex appears to be similar to areas 2 and 3 of the rat as described by Krieg ('46). In both species, layer V is divisible into two zones and there is a thin but distinct layer IV.

In the rostral part of the postsupraorbital I area, peculiar aggregations of small py-

ramidal neurons are present in the superficial zone of layer V which extend onto the medial surface of the hemisphere. In sections cut tangentially with respect to the cortical surface, these clusters have a solid round or oval center surrounded by a less cellular region. Similar clusters were described as being present in layer IV of the neocortex of the brush-tailed possum (Weller, '72) and were designated as "barrels" because of their resemblance to similar, but hollow structures present in the mouse (Woolsey and Van der Loos, '70) and rat (Welker, '71). In these rodents, the barrels corresponded exactly in number to the number of mystacial vibrissae. In the brush-tailed possum, however, there were more barrels than the number of vibrissae over the whole body, but still the barrels were confined to somatic sensory neocortex and were thought to be related in some way to sensory fields (Weller, '72).

In a preliminary report (Royce, '73), it was stated that the aggregations of neurons, termed barrels, in this cortical region were located in layer IV. However, a careful re-evaluation of the sections of all brains in which these structures appear leaves little doubt that they are present in the superficial zone of layer V, and are indeed clusters of small pyramidal neurons. This finding seems reasonable in view of the fact that these clusters are found only in a region which produces movements of the forelimb upon stimulation (Dom et al., '71) and in which we have been unable to record any sensory evoked potentials. Also, the fact that these aggregations extend onto the medial hemisphere seems to correlate with the production of hindlimb movements from electrical stimulation of this region (Dom et al., '71). The possibility that these clusters, even though located in layer V, rather than layer IV, might still in some way be related to sensory hair fields seems unlikely since this species has little hair anywhere except on its abdomen, has no noticeable vibrissae on its snout, and has only a few hairs on the proximal portions of its extremities. It seems more likely that these clusters of cells in layer V are related to the innervation of limb musculature and may serve in some way to increase the efficiency of the very active burrowing and digging activities for which this species is noted (Young, '62).

The cortex of postsupraorbital II appears generally to resemble area 1 of the rat as described by Krieg ('46), in that all six layers are well represented, but layer VI is particularly robust. In contrast to the rat, however, layers II and III are broken up into clusters on the lateral aspect of postsupraorbital II. This lateral region corresponds to the large somatic motor area of the face, tongue and ear (Dom et al., '71) and the partially overlapping sensory area of the face and tongue. The sensory area of the tongue is particularly large and extends rostral to the supraorbital sulcus. The sensory areas for the limbs and trunk are located more dorsally and medially. The function of the cell clusters of layers II and III in this region is unknown.

Our finding that the lateral portion of postsupraorbital II is sensory as well as motor explains why a lesion in this area produced degeneration in the somatosensory thalamus and certain brain stem sensory nuclei as well as within the red nucleus and spinal cord (Harting and Martin, '70a,b). The relationship of the motor and sensory representations in the postsupraorbital area of the armadillo, as defined physiologically, is indicated in the homunculus-type drawing of figure 5. It may be noted that the sensory area for the face and tongue faces rostrally, whereas the motor representation of these regions faces mainly ventrally. This is in contrast to the orientation of the limbs and digits which face rostrally in both the motor and sensory representations. These two homunculi seem to be rotated at an uncertain angle with respect to one another.

The mirror-image relationship of motor to sensory homunculi such as is present in the rat, rabbit, cat, and monkey, seems to be the most common mammalian pattern. In these species, the sensory representation is directed rostrally and the motor representation faces caudally, in addition, the two areas do not overlap, even when mapped by surface recordings (Woolsey, '58). A variation of this pattern is present in the hedgehog in which the motor representation faces caudally, the sensory region faces rostrally, but with considerable overlap in the middle (Lende and Sadler, '67). Previous to this report, the opossum (Lende, '63a,b) and echidna (Lende, '64) were the only species in which the digits of the

primary motor representation are rostral to the limbs and body, however, in these species the primary motor and sensory areas were shown to overlap precisely, point to point. The armadillo is unique among species studied to date in that motor and sensory homunculi are partially separated, but the digits are rostral to the trunk in both representations. Lende ('64) found a separate motor area rostral to motor-sensory amalgam in echidna, but he was unable to determine its orientation.

The armadillo visual cortex appears to be quite different from that of other species. The area from which visual evoked potentials were obtained was small and, at least by our techniques, partially overlapped the caudal portion of the auditory region. Highly visually-oriented animals such as the squirrel and tree shrew possess a distinctive striate cortex with a wide, deeply-staining bilaminar layer IV (Snyder and Diamond, '68; Diamond et al., '70; Hall et al., '71; Kaas et al., '72). This bilaminar feature of layer IV is not present in less visually-oriented species, and it is only the width of layer IV which serves to differentiate striate cortex from other regions in small rodents and bats (Brodmann, '09), the opossum (Gray, '24; Benevento and Ebner, '71), the marsupial phalanger (Martin and Megirian, '72), the rat (Lashley, '34; Krieg, '46), the cat (O'Leary, '41) and the rabbit (Rose and Malis, '65). In the armadillo layer IV is relatively thin (0.4 mm) throughout the visual region, being of similar width to the comparable layer in the motor forelimb region (Postsupraorbital I). The possibility exists that this layer may be functionally wider than apparent because there are granule cells intermixed with pyramidal cells in the superficial zone of layer V and Golgi material reveals that stellate cells, similar to those in the striate area of the cat (O'Leary, '41), and the rat, cat, and monkey (Colonnier, '64) are present in the superficial zone of layer V. The striate cortex of the armadillo is similar in one respect to that of other species in that layers II and III are fused, consisting of an admixture of granular and pyramidal neurons. This condition was also found in the rat (Krieg, '46), rabbit (Rose and Malis, '65) and hedgehog (Hall and Diamond, '68). However, no cytoarchitectural criterion, including the thickness of layer IV and the

fusion of layers II and III proved adequate to define accurately the boundaries of the striate region in the armadillo. The thickness of layer IV remains constant throughout most of the visual area (0.4 mm), but increases gradually (to 0.8 mm) as the auditory region is approached. Also, there are only slight structural differences throughout the visual region of the armadillo and the entire area is most similar to area 18 in the rat (Krieg, '46). No part of it resembles area 17 in the rat or any other species with which we are familiar.

Some difficulties were encountered by Hall and Diamond ('68) in defining the boundaries of the striate cortex in the hedgehog although even in that species there was an increased thickness of layer IV. However, myelin-stained sections showed abundant myelin in the striate area which dropped off sharply on either side of this region. Kaas et al., ('70) recorded visual receptive fields in the hedgehog and found a reversal in the horizontal dimension at the border between areas 17 and 18. Knife cuts were made at this border while recording and myelin-stained sections showed that the cuts were located exactly at the junction of the heavily and lightly myelinated areas. By this technique these authors identified striate and peristriate areas in the hedgehog both physiologically and structurally. A peristriate area has also been demonstrated structurally in the opossum (Gray, '24; Walsh and Ebner, '69; Benevento and Ebner, '71), the marsupial phalanger (Martin and Megirian, '72), the mouse (Valverde and Esteban, '68) and the rat (Krieg, '46), to name a few among the many species in which this area has been demonstrated. On the basis of this evidence we have concluded that the heavily-myelinated area in the visual region of the armadillo is comparable to striate cortex, or area 17, and the lightly stained area lateral to it is the peristriate strip or area 18. Evoked potentials resulting from visual stimulation were recorded throughout this region, but confirmation of the boundaries between areas 17 and 18 in the armadillo awaits receptive field mapping.

Adjacent to the medial boundary of the armadillo striate cortex is a distinctive area characterized primarily by a densely populated layer II. This region is similar to the retrosplenial cortex in the rat (Krieg, '46),

cat (Rose and Malis, '65), and squirrel (Kaas et al., '72). In the armadillo, a sulcus is inconsistently present between the striate and retrosplenial areas which may be sulcus (α) of Smith (1899). In the depths of this sulcus, and on either side of it, is a small transitional region. Rose and Malis ('65) reported that there is a transitional region located in a similar position in the rabbit, which they designated as medial peristriate cortex. Hall et al. ('71) cited these authors and advanced the idea that this transitional region may be a less highly differentiated portion of the striate area rather than medial peristriate cortex. This hypothesis was based on their findings that the lateral well-differentiated striate cortex in the squirrel is concerned with binocular vision, whereas the medial less-differentiated zone receives input only from the contralateral eye. Binocular and unocular cortical visual fields have yet to be investigated in the armadillo and until they are, no definitive statement can be made about the presence or absence of medial peristriate cortex in this species.

The auditory region of the armadillo is less unusual than the striate area in that it shares many characteristics in common with similar regions in other species. Auditory stimulation produced evoked potentials in a relatively large sector of the caudal armadillo neocortex. The auditory area overlapped the somatic sensory areas for the scalp and back rostrally and a portion of the visual area dorsally. Neocortex in the auditory region was designated by the broad term "temporal cortex" which was thought to be preferable at this time to the use of numbers of Brodmann ('09), such as areas 22, 41 and 42. The principal cytoarchitectural features of cortex in the temporal area are: (1) the predominance of granular layers, particularly layer IV which is wide (0.8 mm); (2) the general blurring of lamination and (3) the regularly-spaced aggregations of cells in layers II and III. Except for the last feature, this cortex seems quite similar to the first auditory area in the cat as described by Rose ('49). He reported that there was a fusion of layers II-IV, containing small to medium-sized cells, especially in layer IV, a thin and discontinuous layer III and a small-celled layer V in this region in the cat. Such features are also present in the temporal region of the armadillo. In

contrast, Krieg ('46) reported that the auditory area of the rat, area 41, contained a thin layer IV consisting of small granular cells, and a moderately thick layer V consisting of large and small pyramidal cells. In the temporal (auditory, Lende, '63a) area of the opossum there is a distinct, but thin layer IV and a thick well-developed layer V (Gray, '24). Thus the temporal cortex of the armadillo, in some respects, bears some similarity to the auditory area of the cat, which presumably has a more specialized auditory system, than to that of either the opossum or rat.

Also, in terms of myeloarchitecture, the temporal cortex of the armadillo seems rather advanced. There are distinct inner and outer bands of Baillarger in the temporal region which are not present in the striate area. Similar, distinct bands of Baillarger were also shown to be present in the temporal auditory cortex of the squirrel (Kaas et al., '72). In contrast, photographs of sections in the hedgehog stained for fibers show considerable myelin in the auditory region, but no discernible bands (Hall and Diamond, '68; Kaas et al., '70). Also present in some Weil-stained sections of the armadillo is a peculiar band of heavily-myelinated fibers which runs through layer II. Thus, both physiologically, and anatomically, the visual cortex of the armadillo seems very generalized and the auditory cortex is at least comparable to that of some advanced species. This can be correlated with our incidental observations that the armadillo has small eyes, tiny optic nerves and a non-laminated lateral geniculate body, whereas the ears are large, the auditory nerves are prominent and the medial geniculate body is larger than the lateral geniculate body. Test of visual and auditory acuity, however, are needed to compare the relative functional importance of these systems in this species.

Harting and Martin ('70a) reported that a lesion in the caudal one-third of the neocortex of the armadillo produced degeneration in the pretectal area, superior colliculus and lateral geniculate body. Degeneration from this lesion was also present in the inferior colliculus and medial geniculate body. A smaller lesion of the caudal pole produced degeneration in the lateral geniculate body and superior colliculus, but not in auditory structures. They inter-

preted these findings to mean that temporal fibers were undercut by the larger lesion, and that interpretation would seem to be supported by our findings.

Lateral and ventral to the auditory region in the armadillo the cortex gradually becomes thinner, until at the rhinal fissure it is only 1.5 mm in width. Layer IV gradually becomes diminished or absent, and at the rhinal fissure the only prominent layer is layer VI. Weil-stained sections show few myelinated fibers in this area. This cortex has been designated insular cortex since it is very similar to the ventral insular area in the rat (Krieg, '46), which has few myelinated fibers, and also resembles the insular cortex of the opossum (Gray, '24; Walsh and Ebner, '70). Since this area was silent in response to the stimuli used in this study, little can be said about its functional significance.

In conclusion, it may be stated that the neocortex of the armadillo represents a curious blend of generalized and specialized features. The striate cortex seems generalized in the extreme (closely resembling the peristriate area) although in many respects the motor-sensory and insular regions are comparable to similar areas in other species. Although the auditory cortex, both physiologically and structurally shares features in common with other species, the presence of clusters of neurons in layer V of the rostral motor cortex, the unique relationship of the motor and sensory homunculi, and the presence in lateral regions of cell aggregates in layers II and III are features found in no other species.

ACKNOWLEDGMENTS

The authors express appreciation to Dr. John Harting and Dr. Norman Strominger for their valuable aid and advice as well as to Mr. Terry Stewart and Mr. Gabriel Palukutti for photographic assistance. Gratitude is also extended to Dr. R. W. Guillery for his helpful comments on the manuscript.

This work was supported by Albany Medical College grant GRS-FR-5394 and the University of Wisconsin Graduate School Grant 135-4449 to Dr. G. James Royce and NS-07410 to Dr. George Martin.

LITERATURE CITED

Akert, K. 1964 Comparative anatomy of frontal

- cortex and thalamofrontal connections. In: The Frontal Granular Cortex and Behavior. J. M. Warren and K. Akert, eds. McGraw-Hill Book Co., New York, pp. 372-396.
- Benevento, L. A., and F. F. Ebner 1971 The areas and layers of cortico-cortical terminations in the visual cortex of the Virginia opossum. *J. Comp. Neur.*, 141: 157-190.
- Bodian, D. 1941 Studies on the diencephalon of the Virginia opossum. III. The thalamocortical projection. *J. Comp. Neur.*, 77: 525-576.
- Brodman, K. 1909 Vergleichende Lokalisationstheorie der Grosshirnrinde in ihren Prinzipien dargestellt auf Grund des Zellenbaues. Barth, Leipzig.
- Colonnier, M. 1964 The tangential organization of the visual cortex. *J. Anat.*, (London), 98: 327-344.
- Diamond, I. T., M. Snyder, H. Killackey, J. Jane and W. C. Hall 1970 Thalamo-cortical projections in the tree shrew (*Tupaia glis*). *J. Comp. Neur.*, 139: 273-306.
- Dom, R., G. F. Martin, B. L. Fisher, A. M. Fisher and J. K. Harting 1971 The motor cortex and corticospinal tract of the armadillo (*Dasypus novemcinctus*). *J. Neurol. Sci.*, 14: 225-236.
- Fisher, A. M., J. K. Harting, G. F. Martin and M. I. Stuber 1969 The origin, course and termination of corticospinal fibers in the armadillo (*Dasypus novemcinctus mexicanus*). *J. Neurol. Sci.*, 8: 347-361.
- Fox, C. A., M. Ubeda-Purkiss, H. K. Ihrig and D. Biagioli 1951 Zinc chromate modification of the Golgi technic. *Stain Tech.*, 26: 109-114.
- Gray, P. A. 1924 The cortical lamination pattern of the opossum, *Didelphis virginiana*. *J. Comp. Neur.*, 37: 221-263.
- Hall, W. C., and I. T. Diamond 1968 Organization and function of the visual cortex in hedgehog: I. Cortical cytoarchitecture and thalamic retrograde degeneration. *Brain, Behav. Evol.*, 1: 181-214.
- Hall, W. C., J. H. Kaas, H. Killackey and I. T. Diamond 1971 Cortical visual areas in the grey squirrel (*Sciurus carolinensis*): a correlation between cortical evoked potential maps and architectonic subdivisions. *J. Neurophysiol.*, 34: 437-452.
- Harting, J. K., and G. F. Martin 1970a Neocortical projections to the mesencephalon of the armadillo, *Dasypus novemcinctus*. *Brain Res.*, 17: 447-462.
- 1970b Neocortical projections to the pons and medulla of the nine-banded armadillo (*Dasypus novemcinctus*). *J. Comp. Neur.*, 138: 483-500.
- Kaas, J., W. C. Hall and I. T. Diamond 1970 Cortical visual areas I and II in the hedgehog: relation between evoked potential maps and architectonic subdivisions. *J. Neurophysiol.*, 33: 595-615.
- 1972 Visual cortex of the grey squirrel (*Sciurus carolinensis*): architectonic subdivisions and connections from the visual thalamus. *J. Comp. Neur.*, 145: 273-306.
- Krieg, W. J. S. 1946 Connections of the cerebral cortex. I. The albino rat. B. Structure of the cortical areas. *J. Comp. Neur.*, 84: 277-323.
- Lashley, K. S. 1934 The mechanism of vision. VIII. The projection of the retina upon the cerebral cortex of the rat. *J. Comp. Neur.*, 60: 57-79.

- Lende, R. A. 1963a Sensory representation in the cerebral cortex of the opossum (*Didelphis virginiana*). *J. Comp. Neur.*, 121: 395-404.
- 1963b Motor representation in the cerebral cortex of the opossum (*Didelphis virginiana*). *J. Comp. Neur.*, 121: 405-415.
- 1964 Representation in the cerebral cortex of a primitive mammal. Sensorimotor visual and auditory fields in the echidna (*Tachyglossus aculeatus*). *J. Neurophysiol.*, 27: 34-48.
- Lende, R. A., and K. M. Sadler 1967 Sensory and motor areas in neocortex of hedgehog (*Eri-naceus*). *Brain Res.*, 5: 390-405.
- Martin, G. F. 1968 The pattern of neocortical projections to the mesencephalon of the opossum, *Didelphis virginiana*. *Brain Res.*, 11: 593-610.
- Martin, G. F., and A. M. Fisher 1968 A further evaluation of the origin, the course and the termination of the opossum corticospinal tract. *J. Neurol. Sci.*, 7: 177-187.
- Martin, G. F., and D. Megirian 1972 Corticobulbar projections of the marsupial phalanger (*Trichosurus vulpecula*). II. Projections to the mesencephalon. *J. Comp. Neur.*, 144: 165-192.
- Morest, D. K., and R. R. Morest 1966 Perfusion-fixation of the brain with chrome-osmium solutions for the rapid Golgi method. *Am. J. Anat.*, 118: 832.
- O'Leary, J. L. 1941 Structure of the area striata of the cat. *J. Comp. Neur.*, 75: 131-164.
- Pubols, B. H. 1968 Retrograde degeneration study of somatic sensory thalamocortical connections in brain of virginia opossum. *Brain Res.*, 7: 232-251.
- Ramón-Moliner, E. 1970 The Golgi-Cox technique. In: *Contemporary Research Methods in Neuroanatomy*. W. J. H. Nauta and S. O. E. Ebbesson, eds. Springer-Verlag, New York, pp. 32-55.
- Rose, J. E. 1949 The cellular structure of the auditory region of the cat. *J. Comp. Neur.*, 91: 409-440.
- Rose, J. E., and C. N. Woolsey 1948 The orbito-frontal cortex and its connections with the medio-dorsal nucleus in the rabbit, sheep, and cat. In: *The Frontal Lobes* (Res. Publ. Ass. Nerv. Ment. Dis., vol. 27). J. F. Fulton, C. D. Aring and S. B. Wortis, eds. Williams and Wilkins, Baltimore, pp. 210-232.
- Rose, J. E., and L. I. Malis 1965 Geniculo-striate connections in the rabbit. I. Cytoarchitectonic structure of the striate region and of the dorsal lateral geniculate body; organization of the geniculostriate projections. *J. Comp. Neur.*, 125: 121-140.
- Royce, G. J. 1973 Neocortical architecture and functional localization in the nine-banded armadillo (*Dasypus novemcinctus*). *Anat. Rec.*, 175: 431.
- Sholl, D. A. 1953 Dendritic organization in the neurons of the visual and motor cortices of the cat. *J. Anat.*, (London), 87: 387-406.
- Smith, G. E. 1899 The brain in the Edentata. *Proc. Linnean Soc. Second Series*, 7: 277-394.
- Snyder, M., and I. T. Diamond 1968 The organization and function of the visual cortex in the tree shrew. *Brain, Behav. Evol.*, 1: 244-288.
- Strominger, N. L. 1969 A comparison of the pyramidal tracts in two species of edentate. *Brain Res.*, 15: 259-262.
- Valverde, F., and M. E. Esteban 1968 Peristriate cortex of mouse: location and the effects of enucleation on the number of dendritic spines. *Brain Res.*, 9: 145-148.
- Voneida, T. J., and G. J. Royce 1974 Ipsilateral connections of the gyrus proreus in the cat. *Brain Res.*, 76: 393-400.
- Walsh, T. M., and F. F. Ebner 1970 The cyto-architecture of somatic sensory-motor cortex in the opossum (*Didelphis marsupialis virginiana*): a Golgi study. *J. Anat.*, (London), 107: 1-18.
- Welker, C. 1971 Microelectrode delineation of fine grain somatotopic organization of Sm I cerebral neocortex in albino rat. *Brain Res.*, 26: 259-275.
- Weller, W. L. 1972 Barrels in somatic sensory neocortex of the marsupial *Trichosurus vulpecula* (brush-tailed possum). *Brain Res.*, 43: 11-24.
- Woolsey, C. N. 1958 Organization of somatic sensory and motor areas of the cerebral cortex. In: *Biological and Biochemical Basis of Behavior*. H. H. Harlow and C. N. Woolsey, eds. The University of Wisconsin Press, Madison, pp. 63-81.
- Woolsey, T. A., and H. Van der Loos 1970 The structural organization of layer IV in the somato-sensory region (SI) in mouse cerebral cortex. *Brain Res.*, 17: 205-242.
- Young, J. Z. 1962 *The Life of Vertebrates*. Oxford University Press, New York.

PLATES

PLATE 1

EXPLANATION OF FIGURES

- 8 Low power photomicrograph of a thionin stained frontal section through the presupraorbital cortex (PRS) and the rostral portion of the postsupraorbital I cortex (PST I). The dashed lines in the presupraorbital region outline the approximate region shown in higher power in figure 9; whereas, the dashed lines in the postsupraorbital I cortex demarcate the approximate region shown at higher power in figure 10. The supraorbital sulcus (SO), rhinal fissure (RF), and part of the pyriform cortex (PYR) are also shown.
- 9 High power photomicrograph of the portion of the presupraorbital cortex outlined in figure 8. The cortical layers are indicated by Roman numerals.
- 10 High power photomicrograph of the portion of the postsupraorbital I cortex outlined in figure 8. The cortical layers are indicated by Roman numerals and the area in the rectangle is the approximate region shown at higher power in figure 11.
- 11 Higher power photomicrograph of the portion of the postsupraorbital I cortex outlined in figure 10. The cortical layers are indicated on the left by Roman numerals. The aggregate of neurons in the center of the photomicrograph consists of cells of layer Va, and is similar in appearance to the "barrels" present in layer IV in some other species.

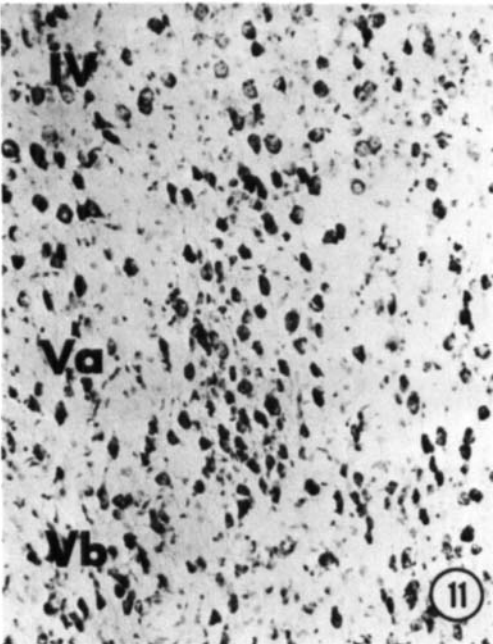
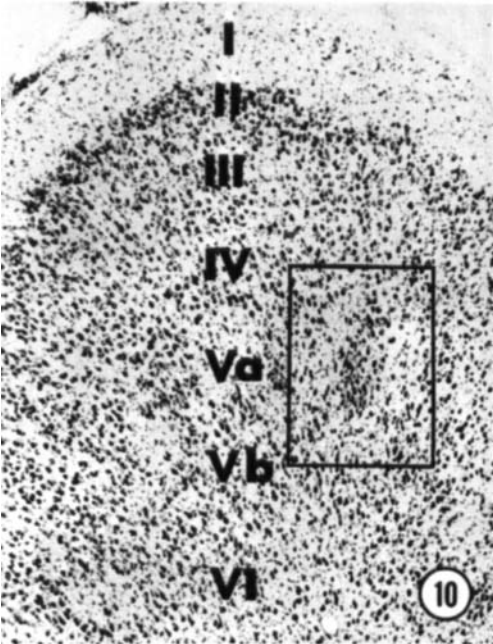
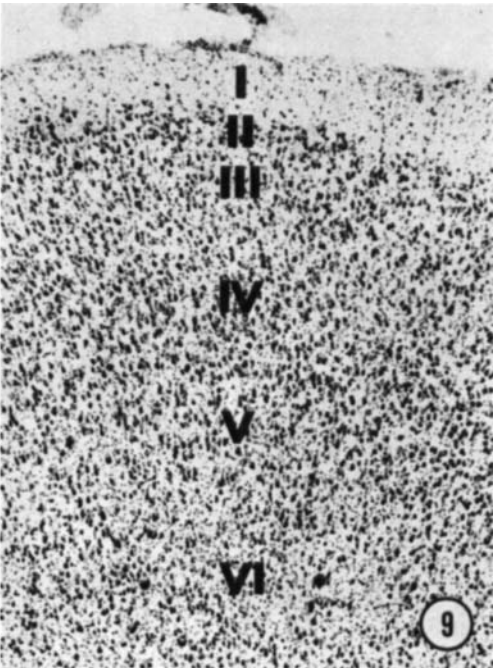
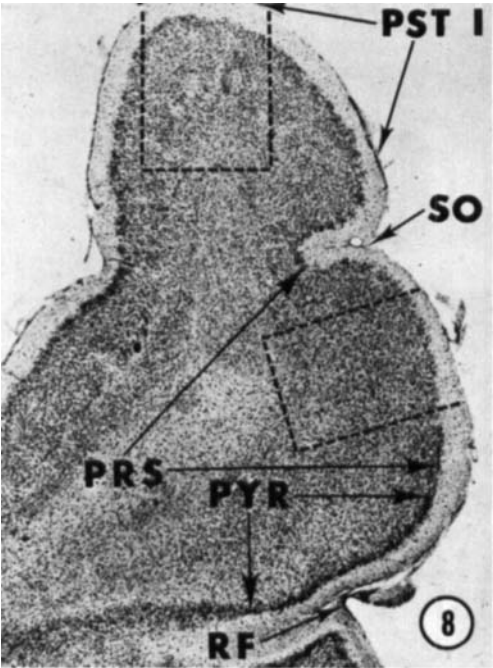


PLATE 2

EXPLANATION OF FIGURES

- 12 Low power photomicrograph of a thionin stained frontal section through the middle of the sagittal sulcus (SG). Medial to the sulcus is the neocortex of postsupraorbital I and lateral to it is the neocortex of postsupraorbital II. The dashed lines in postsupraorbital I outlines the approximate region shown at higher power in figure 13; whereas the dashed lines in the cortex of postsupraorbital II indicate the approximate region shown at higher power in figure 15. The position of the rhinal fissure (RF) and the pyriform cortex (PYR) are indicated for reference.
- 13 High power photomicrograph of the portion of the cortex of postsupraorbital I outlined in figure 12. The cortical layers are indicated by Roman numerals.
- 14 Higher power photomicrograph of a neuron from layer Vb of the cortex of postsupraorbital I in a 30-day old armadillo impregnated by the Golgi-Cox method and counterstained with cresyl violet.
- 15 Higher power photomicrograph of the portion of the cortex of postsupraorbital II outlined in figure 12. The cortical layers are indicated by Roman numerals.

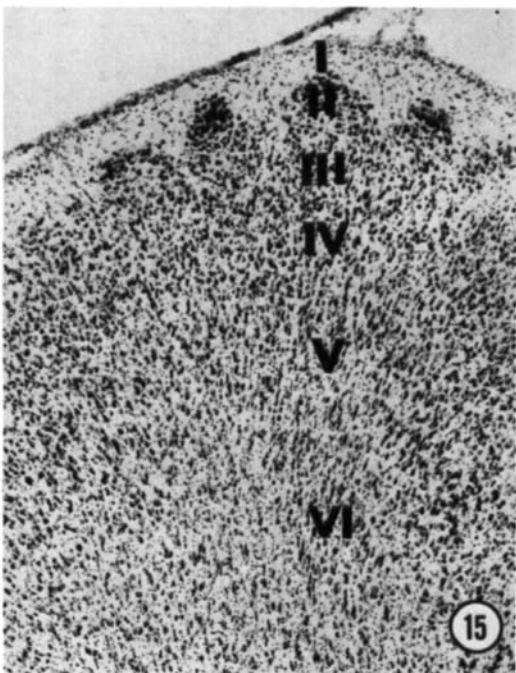
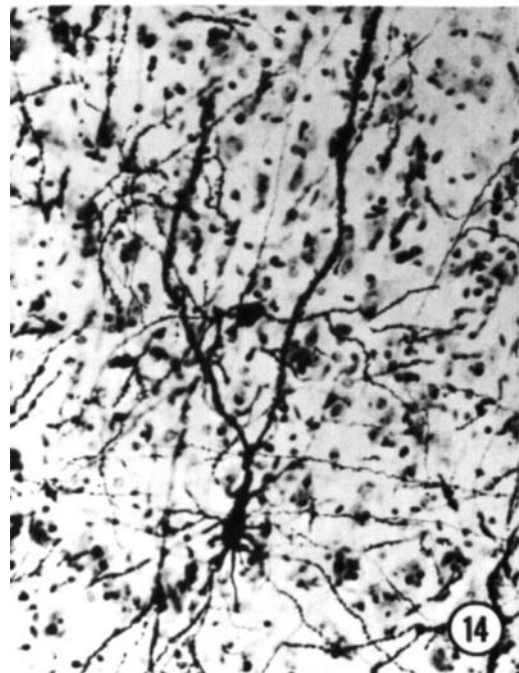
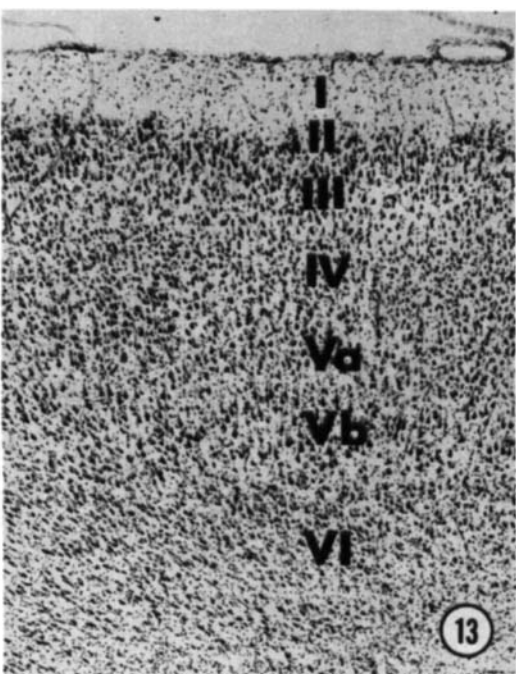
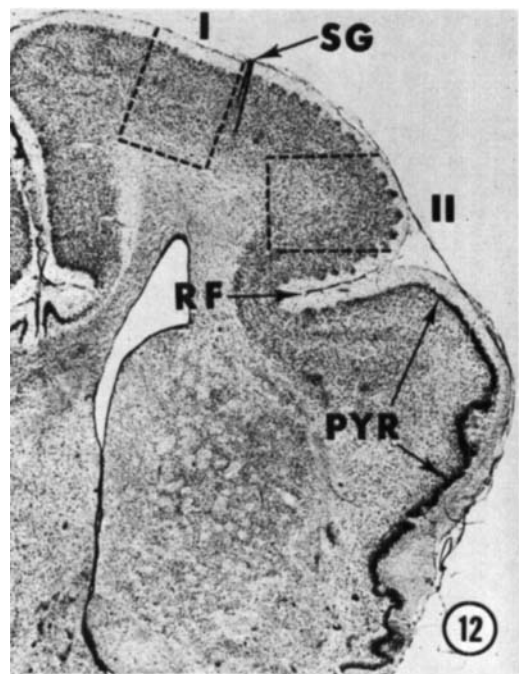


PLATE 3

EXPLANATION OF FIGURE

- 16 Low power photomicrograph of a section located in the region of the sagittal sulcus (SG), from an adult armadillo impregnated by the Golgi-Cox method. The cortex of postsupraorbital I is medial to the sulcus, and lateral to the sulcus is the cortex of postsupraorbital II.

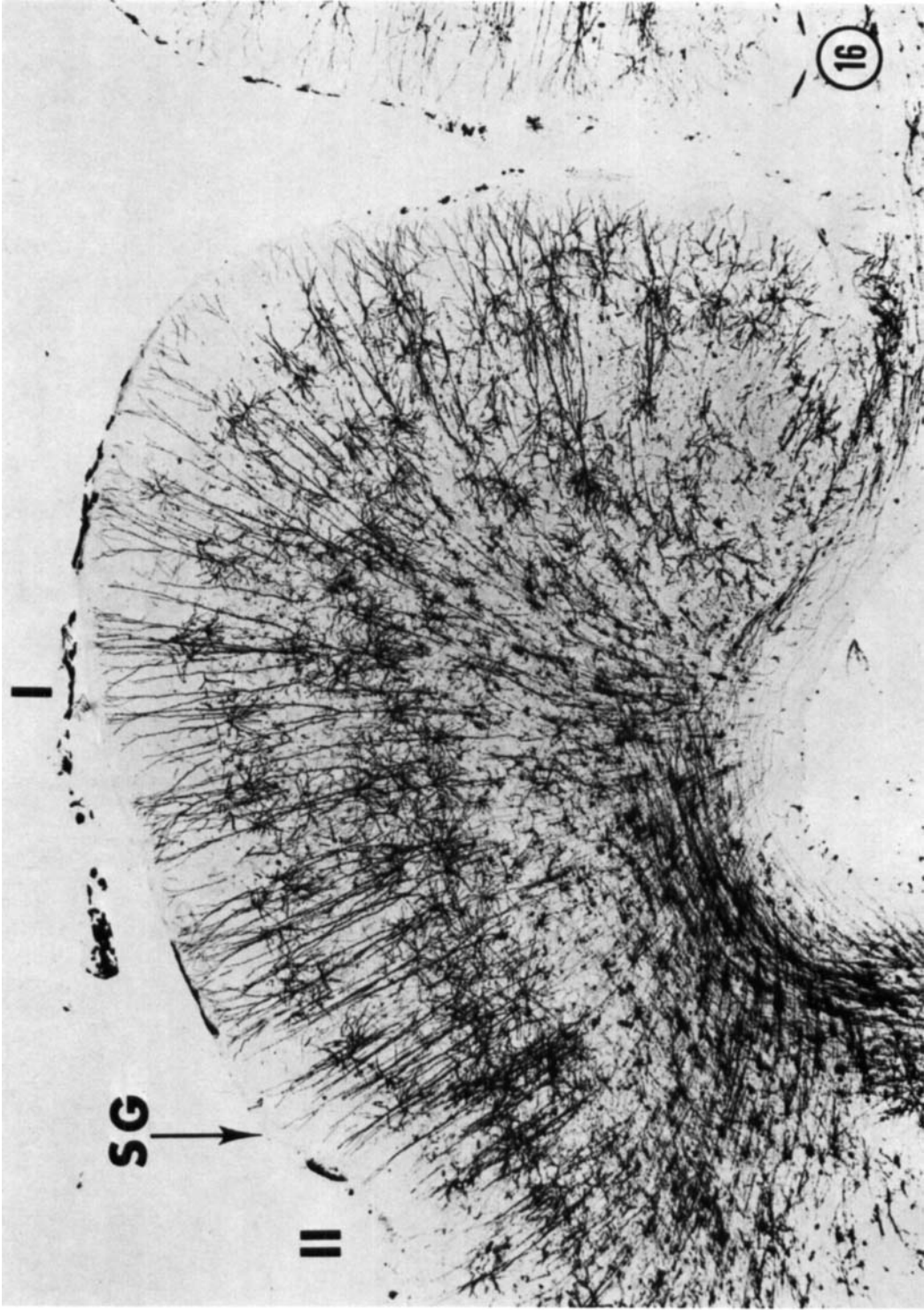


PLATE 4

EXPLANATION OF FIGURES

- 17 Low power photomicrograph of a thionin stained frontal section through the middle of the striate cortex. The cortex between the most dorsal set of arrows is the striate or visual cortex (Sv), and more ventrally the arrows bound the peristriate cortex (PS), the temporal or auditory cortex (Ta) and the insular cortex (IN), respectively, the retrosplenial area (RS), and the pyriform cortex (PYR) are indicated for reference. The asterisk (*) indicates the location of the rhinal fissure. The dashed lines in the temporal area indicate the approximate region shown at higher power in figure 18. The dashed lines in the striate and the peristriate areas indicate the approximate region shown at higher power in figure 19 and figure 20, respectively. The dashed lines in the area of the insular cortex and rhinal fissure indicate the approximate region shown at higher power in figure 24.
- 18 High power photomicrograph of the portion of the temporal cortex outlined in figure 17. The cortical layers are indicated by Roman numerals.
- 19 High power photomicrograph of the portion of the striate cortex outlined in figure 17. The cortical layers are indicated by Roman numerals.
- 20 High power photomicrograph of the portion of the peristriate cortex outlined in figure 17. The cortical layers are indicated by Roman numerals.

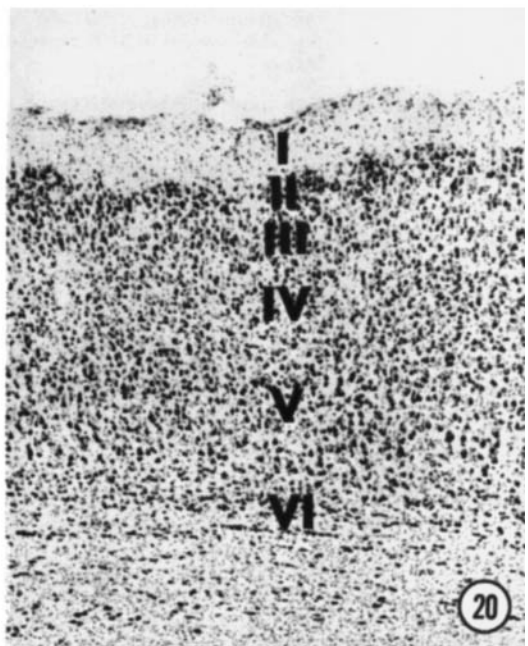
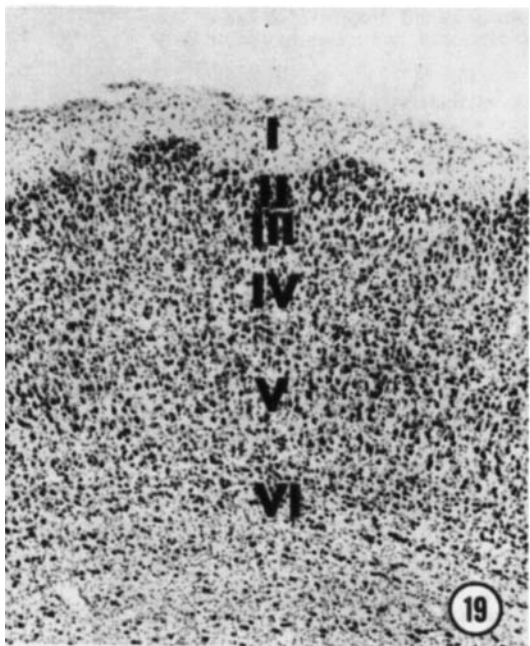
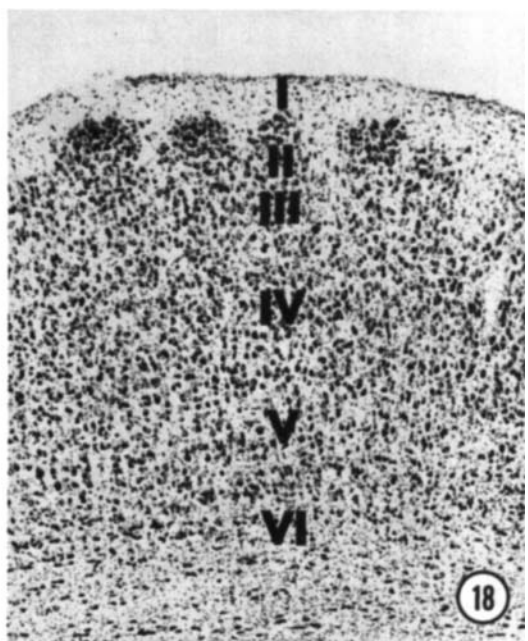
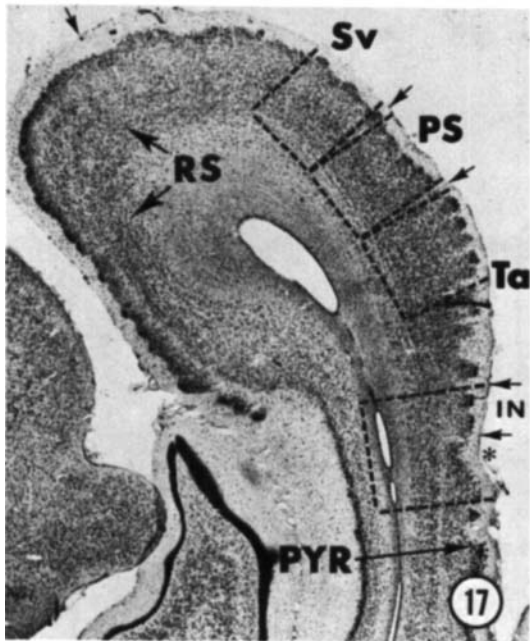


PLATE 5

EXPLANATION OF FIGURES

- 21 Higher power photomicrograph of a Golgi-Cox impregnated section in the striate cortex of an adult armadillo. Stellate or star cells are prominent in layer IV (top), but are also present to a lesser extent in the superficial zone of layer V (bottom).
- 22 Low power photomicrograph of a Weil-stained section through the caudal neocortex. Arrows bound the regions of the striate or visual cortex (Sv), the peristriate cortex (PS), the temporal or auditory cortex (Ta), and the insular cortex (IN). The retrosplenial cortex (RS), and the pyriform cortex (PYR) are indicated for reference. The rhinal fissure is indicated by an asterisk (*). Note the homogeneous nature of the fibers in the striate region compared to the laminated pattern of fibers in the temporal region which represents the inner and outer bands of Bailarger.
- 23 Low power photomicrograph of a Weil-stained section through the temporal cortex at a level rostral to that shown in figure 22. The arrow points to a thin band of heavily myelinated fibers which is present in layer II on the lateral aspect of the temporal or auditory (Ta) region. The position of the rhinal fissure (RF) is indicated for reference.
- 24 High power photomicrograph of the portion of the insular cortex (IN) on the left and the region of the rhinal fissure (RF) outlined in figure 17. The cortical layers of the insular cortex are indicated on the left by Roman numerals.

